

From Legal PDFs to Verifiable Answers

Evaluating answer quality, page grounding, valid output, and latency across 100 legal questions

01

Hypothesis

Can a Legal RAG pipeline balance answer quality, grounding, valid output, and latency while preserving auditable page provenance?
H1 · System profile. Targets: answer ≥ 0.90 ; grounding ≥ 0.85 ; 100% valid; every annotated question retains a correct page; latency multiplier ≥ 0.90 .
H2 · Attribution. Select pages after answering to raise macro precision, reduce citation size, and retain a correct page for all 95 annotated questions.

02

Methodology · one auditable chain

01

Legal PDFs

30 public files · 590 pages

→

02

Page-stable corpus

Text · tables · OCR fallback

→

03

Multi-view index

Page · section · clause · table

→

04

Hybrid retrieval

BM25 · dense · RRF · rerank

→

05

Typed answer

Strict JSON · support checks

→

06

Attribution + output

Pages · trace · telemetry

CONTINUOUS PROVENANCE · document_id + page_number

Data

100 questions · 95 page-annotated

Evaluation

paired stages · $\beta = 2.5$

Models

Qwen3 Embedding/Reranker 0.6B · gpt-5.4-mini

03

Results

SYSTEM · 100 ANSWERS

Grounding and latency constrain strong answers

H1 is partially supported: grounding 0.847 falls 0.003 below the retrospective 0.85 target.

0.956 × 0.847 × 1.000 × 0.938 = 0.759

answer quality benchmark grounding valid telemetry latency multiplier total score

Multiplicative contract: strong answer quality is discounted by grounding and latency.

100 / 100 valid

All outputs passed schema checks.

TTFT 6.39 s

Mean time to first token.

ATTRIBUTION · 95 ANNOTATED QUESTIONS

Answer-aware selection removes noise, not every risk

Macro page metrics on 95 annotated questions: the final 89.2% F β is separate from benchmark grounding 0.847 on all 100 answers.

	RAW		FILTER		FINAL
PRECISION	63.8%	→	63.8%	→	73.9%
RECALL	96.8%	→	96.2%	→	95.2%
MACRO PAGE F β $\beta = 2.5$ · n = 95	87.7%	→	87.3%	→	89.2%

+10.2 pp

precision · 95% CI +6.3 to +14.4

-0.33

pages per question · 2.56 → 2.23

95 / 95

retain ≥ 1 correct page

BOUNDARY · EXPLORATORY SPLIT

Narrowing helps local evidence and harms distributed evidence

Raw-to-final change in page F β . Blue improves; amber regresses. Sample sizes sum to the same 95 questions.

SINGLE PAGE · n=59

MULTI PAGE · n=36

SINGLE DOC · n=60

MULTI DOC · n=35

-4

0 pp

+4

4.0 pp

-2.6 pp

+3.2 pp

-1.4 pp

04 · DECISION

Engineering rule

Retrieve broadly. Attribute explicitly. Narrow conservatively when support spans pages or documents.

Limitations

- One saved run; retrospective thresholds; no component ablation.
- One DIFC benchmark; OCR fallback was implemented but not exercised.
- Only TTFT was measured - not full latency, throughput, cost, or energy.